

Regulation of Yeast Pyruvate Kinase by Ultrasensitive Allostery Independent of Phosphorylation

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SUMMARY

Allostery and covalent modification are major means of fast-acting metabolic regulation. Their relative roles in responding to environmental changes remain, however, unclear. Here we examine this issue, using as a case study the rapid decrease in pyruvate kinase flux in yeast upon glucose removal. The main pyruvate kinase isozyme (Cdc19) is phosphorylated in response to environmental cues. It also exhibits positively cooperative (ultrasensitive) allosteric activation by fructose-1,6-bisphosphate (FBP). Glucose removal causes accumulation of Cdc19's substrate, phosphoenolpyruvate. This response is retained in strains with altered protein-kinase-A or AMP-activated-protein-kinase activity or with *CDC19* carrying mutated phosphorylation sites. In contrast, yeast engineered with a *CDC19* point mutation that ablates FBP-based regulation fail to accumulate phosphoenolpyruvate. They also fail to grow on ethanol and slowly resume growth upon glucose upshift. Thus, while yeast pyruvate kinase is covalently modified in response to glucose availability, its activity is controlled almost exclusively by ultrasensitive allostery.

INTRODUCTION

The major mechanisms of rapid metabolic regulation are allostery and covalent modification. Allosteric regulation is intrinsic to the metabolic network, enabling automatic adjustment of fluxes in response to metabolite concentration changes. Covalent modification involves distinct signaling enzymes, which integrate metabolism with growth and other cellular functions. The most thoroughly studied form of enzyme covalent modification is protein kinase-catalyzed phosphorylation. Both central metabolic enzymes and major metabolic regulatory kinases are conserved across a wide range of eukaryotes. For example, yeast

and mammals encode the same set of glycolytic and citric acid cycle enzymes, as well as protein kinase A (PKA), protein kinase B (Akt in mammals, Sch9 in yeast), and AMP-activated protein kinase (AMPK in mammals, SNF1 in yeast).

Glycolysis is the metabolic pathway typically exhibiting the highest flux in both yeast and mammals. A common assumption, although subject to some dispute, is that the committed step, catalyzed by phosphofructokinase-1 (Pfk1), functions as the critical glycolytic control point (Galazzo and Bailey, 1990; Moreno-Sánchez et al., 2008; Small and Kacser, 1993; Torres et al., 1986). Pfk1 is stimulated by fructose-2,6-bisphosphate, produced by phosphofructokinase-2 (Pfk2). The activity of Pfk2 is in turn regulated by a kinase cascade. In yeast, PKA phosphorylates Pfk2, elevating fructose-2,6-bisphosphate levels and thereby enhancing Pfk1 activity and glycolytic flux (Kretschmer and Fraenkel, 1991). This regulation is physiologically relevant in yeast during the shift from respiratory to fermentative conditions in order to more rapidly initiate glycolysis when glucose suddenly becomes available (Boles et al., 1996; Raamsdonk et al., 2001). Thus, kinase-based signaling and allostery work in concert to enable rapid regulation of phosphofructokinase.

The terminal step in glycolysis, catalyzed by pyruvate kinase, is also irreversible and tightly regulated. Humans encode four isozymes of pyruvate kinase, one of which (PKM2) plays a central role in oncogenesis (Christofk et al., 2008). Yeast encode two isozymes, Cdc19 and Pyk2, with Cdc19 catalyzing most of the flux and therefore essential for fermentative growth (Boles et al., 1997). The gene name *CDC19* reflects the fact that temperature-sensitive *cdc19* mutants arrest growth in G1 during growth on glucose, for as yet unknown reasons.

Both phosphorylation and allostery contribute to pyruvate kinase regulation. El-Maghrabi et al. (1982) reported that PKA deactivates liver pyruvate kinase, while several groups have reported that PKA activates Cdc19 in yeast by phosphorylating serine 22 (Galello et al., 2010; Portela et al., 2002, 2006; Ptacek et al., 2005). In addition, most eukaryotic pyruvate kinases, including Cdc19, are allosterically activated by the product of phosphofructokinase, fructose-1,6-bisphosphate (FBP) (Blair and Walker, 1984; Dombrackas et al., 2005; Jurica et al., 1998).

During gluconeogenic growth in yeast, Cdc19 is present but inactive (Cortassa et al., 1995; Wilson and Bhattacharjee,

1986). This enzyme inactivation limits futile cycling during steady gluconeogenic growth. To investigate pyruvate kinase regulation further, we have characterized the response of yeast metabolome to sudden changes of its nutrient environment. Short-term nitrogen or phosphate starvation has essentially no effect on the concentrations of glycolytic metabolites. However, glucose removal causes a dramatic accumulation of pyruvate kinase's substrate, phosphoenolpyruvate (PEP). This observation suggests that yeast can rapidly suppress Cdc19 enzyme activity upon glucose deprivation. Here we examine how this inactivation is achieved, with an eye toward potential general lessons regarding enzyme regulation.

One mechanism that could account for the rapid inactivation of Cdc19 after glucose depletion involves PKA inactivation. Another involves depletion of FBP. To distinguish between these possibilities, we use a combination of metabolomic, genetic, and chemical genetic approaches. We demonstrate that inhibition of PKA selectively alters trehalose and cyclic-AMP (cAMP) levels while not altering the levels of other metabolites, including PEP. Instead, we find that PEP levels are controlled by positively cooperative allosteric activation of Cdc19 by FBP. We show that this allosteric regulation is required for gluconeogenic growth. In addition, it contributes to energy charge homeostasis upon glucose upshift: accumulation of lower glycolytic intermediates upon glucose removal provides a readily accessible reservoir of high-energy phosphate bonds that can be used to enable rapid phosphorylation of glucose when it becomes available. Such rapid phosphorylation primes glycolysis on glucose re-addition and contributes to efficient growth in an oscillating glucose environment. These results support a primary role in microbial metabolic flux regulation for allosteric mechanisms intrinsic to the metabolic network itself and raise the possibility that many cases of enzyme covalent modification may be functionally irrelevant.

RESULTS

Glucose Removal Results in PEP Accumulation due to Pyruvate Kinase Inactivation

We used liquid chromatography-mass spectrometry (LC-MS) to measure short-term changes in the metabolome profile of *S. cerevisiae* cells subjected to rapid transition from minimal media to media lacking glucose, nitrogen, or phosphate. For glucose removal, yeast cells were grown either in liquid batch cultures or on filters residing on the surface of an agarose plate. The filter culture technique enables rapid, nondisruptive nutrient removal and metabolome quenching. However, this method was unreliable for experiments using ATP analogs to control genetically engineered kinases, due to variable diffusion of analogs to the cells. Moreover, the filter culture technique retains trace amounts of nutrient after transfer to nutrient-free media. For example, the metabolome pattern after glucose removal using filter culture closely mimics that of a downshift in liquid culture to ~0.01% glucose (Figure S1 available online). For nitrogen and phosphate starvation, only liquid cultures were examined and these were grown in nitrogen or phosphate limited media before nutrient deprivation, to deplete the otherwise large intracellular store of these two nutrients.

Irrespective of the culturing method applied, within 30 min of glucose removal, FBP and dihydroxyacetone phosphate (DHAP) decreased by more than 15-fold, and PEP increased by at least 10-fold (Figure 1A). In contrast, nitrogen or phosphate starvation failed to alter the concentrations of glycolytic metabolites (Figure 1B). Thus, we focused on the metabolic regulatory events induced by glucose removal. The increase in PEP during glucose removal suggests that cells either rapidly suppress PEP consumption or rapidly induce gluconeogenesis after glucose deprivation. To evaluate the possibility that the increased PEP pool results from gluconeogenesis, we switched cells from unlabeled glucose to U-¹³C-ethanol. If PEP were being produced by gluconeogenesis, we would expect incorporation of ¹³C into PEP in this experiment. However, we observed no detectable PEP labeling (<1%) over the first 2 hr after glucose removal, even though a significant amount of labeled malate (>80%) accumulated during this time, documenting utilization of the labeled ethanol (Figure 2). This result rules out substantial PEP formation by PEP carboxykinase in the gluconeogenesis pathway on this time scale, leading us to conclude that PEP consumption is curtailed.

Only two known reactions in *S. cerevisiae* consume PEP: pyruvate kinase and aromatic amino acid biosynthesis, with the former contributing ~99.5% of the total PEP consumption flux in growing cells (Blank et al., 2005). Moreover, because expression of *PYK2* is repressed during growth on glucose and *Pyk2* does not by itself support growth on glucose (Boles et al., 1997), regulation of flux from PEP upon glucose removal must be exerted through the major pyruvate kinase isozyme, Cdc19. The overall Cdc19 concentration did not change upon glucose removal (relative abundance 15 min after glucose removal is 0.97 ± 0.09 of initial abundance, as measured by LC-MS/MS). Thus, Cdc19 enzymatic activity must be regulated.

Glucose-Sensing Kinases Are Not Required for Pyruvate Kinase Regulation

PKA plays a critical role in the response of yeast cells to glucose availability (Zaman et al., 2008). Glucose induces production of cAMP via Ras and Gpa2 signaling pathways, which in turn activate PKA. Upon glucose removal, cAMP levels decrease and PKA is quickly inactivated, resulting in changes in the phosphorylation state of a variety of metabolic enzymes (Zaman et al., 2008). To explore the role of PKA in direct metabolic regulation, we used a yeast strain with genetic modifications to all PKA isozymes (*tpk1*^{M164G}*tpk2*^{M147G}*tpk3*^{M165G}) that result in their specific inhibition by the ATP analog 1NM-PP1. We refer to this strain as the PKA analog-sensitive strain (*pka^{as}*) (Zaman et al., 2009). Consistent with prior literature, analog addition to the *pka^{as}* yeast strain induced greater than 10-fold changes in roughly half of transcripts, with stress response and catabolic gene transcripts increasing and ribosomal ones decreasing (Figure S3A). In addition, growth slowed by ~10% (Figure S2A).

Metabolomic analysis of the *pka^{as}* strain, before and after analog addition, revealed that out of 133 measured metabolites only two showed a substantial concentration changes upon acute PKA inactivation: cAMP and trehalose (Figure 3B). PKA has been proposed to enhance cAMP degradation by activating either or both phosphodiesterases, Pde1 and Pde2 (Zaman

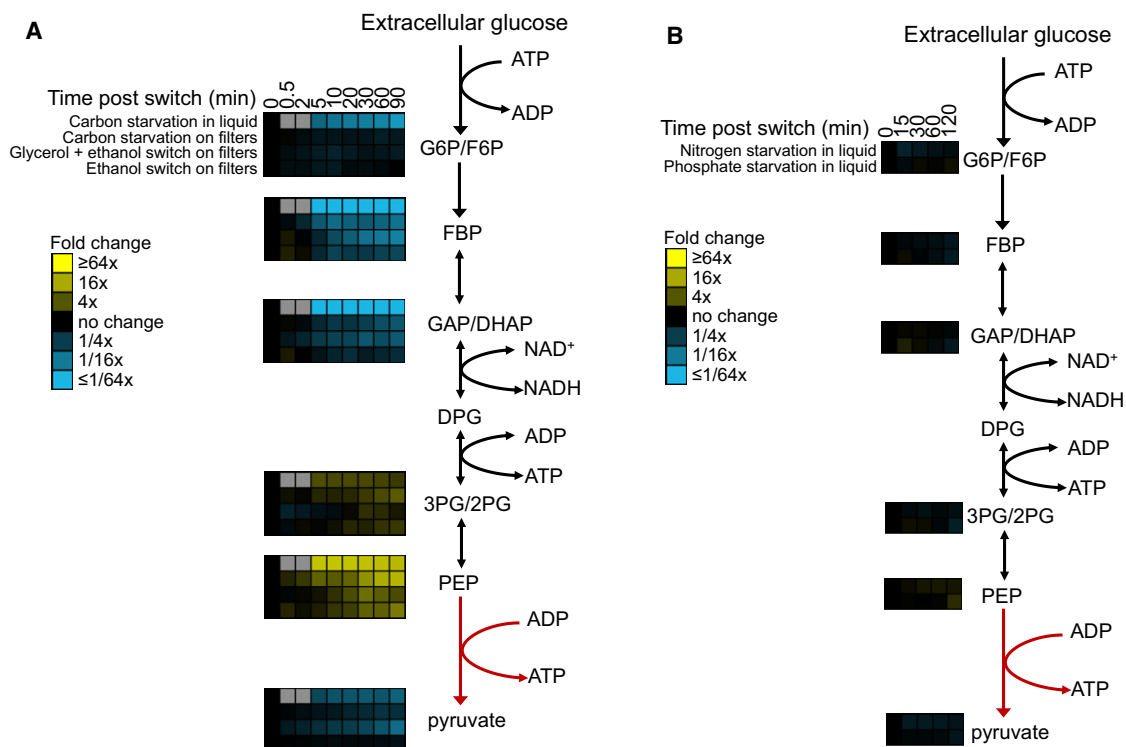


Figure 1. Glucose Removal Results in PEP Accumulation

(A) *S. cerevisiae* cells (strain Y3358) growing in minimal media containing 2% glucose were switched to minimal media containing no carbon, 2% glycerol + 2% ethanol, or 2% ethanol. After the indicated duration of glucose removal, the metabolome was quantified by LC-MS. Experiments were performed with either liquid culture (more complete glucose removal) or filter culture (faster sampling).

(B) *S. cerevisiae* cells (strain Y3358) growing in minimal media were switched to minimal media containing no nitrogen or no phosphate, and metabolome was quantitated by LC-MS. Experiments were performed with liquid culture.

In (A) and (B), data are shown in heatmap format, with each line reflecting the dynamics of a particular compound in a particular culture condition. Metabolite levels of biological duplicates were averaged and normalized to cells growing steadily in glucose (time zero), and the resulting fold changes log transformed. Gray boxes indicate sample not available. G6P, glucose-6-phosphate; F6P, fructose-6-phosphate; FBP, fructose-1,6-bisphosphate; GAP, glyceraldehyde-3-phosphate; DHAP, dihydroxyacetone phosphate; DPG, 1,3-diphosphoglycerate; 3PG, 3-phosphoglycerate; 2PG, 2-phosphoglycerate; G6P and F6P, GAP and DHAP, and 3PG and 2PG were not differentiated by the LC-MS method employed.

See also Figure S1.

et al., 2008). This regulation may provide feedback inhibition of the cAMP-PKA signaling circuit. Consistent with this hypothesis, we find that PKA inactivation leads to dramatically higher cAMP levels. PKA has also been shown to inhibit trehalose-6-phosphate phosphatase, whose product is trehalose, and to activate trehalase, which converts trehalose into glucose (Argüelles et al., 1993; De Mesquita et al., 2003; François et al., 1991). Such regulation would predict that inhibition of PKA should lead to accumulation of trehalose, as we observe. Importantly, our results demonstrate that PKA inhibition not only promotes higher trehalose, but is alone sufficient to produce marked trehalose accumulation. The effects of PKA inhibition on cAMP and trehalose levels document a role for PKA in direct metabolic regulation but suggest that the effect is limited to a few soluble metabolites.

Since Cdc19 activity has been reported to be activated by phosphorylation via PKA both in vitro (Portela et al., 2002; Portela et al., 2006) and in vivo (Rayner et al., 2002), we hypothesized that PKA inactivation would decrease pyruvate kinase activity. However, analog addition did not lead to a detectable elevation

in PEP levels, arguing against a major role for PKA-based regulation of pyruvate kinase activity (Figure 3). Thus, while inactivation of PKA in vivo does yield increased cAMP and trehalose levels, it does not significantly affect Cdc19 activity.

Sch9 shares some targets with PKA and can partially compensate for PKA inactivation (Zaman et al., 2008). It is thus possible that persistent Sch9 activity masked a role for PKA in acute regulation of pyruvate kinase activity. To rule out such a possibility, we used a yeast strain, *pka^{as}sch9^{as}*, in which both PKA and Sch9 were genetically modified (*sch9^{T492G}tpk1^{M164G}tpk2^{M147G}tpk3^{M165G}*) such that they are specifically inhibited by the ATP analog 1NM-PP1. Treatment of this strain with 1NM-PP1 resulted in almost identical metabolome changes as those observed for the *pka^{as}* strain (Figure S2B). Thus, in cells growing in glucose, cAMP and trehalose are the only observed metabolites whose levels are strongly and acutely regulated by PKA/Sch9 inhibition.

Our results with the *pka^{as}* and *pka^{as}sch9^{as}* strains argue that PKA inactivation is not sufficient to substantially inhibit pyruvate

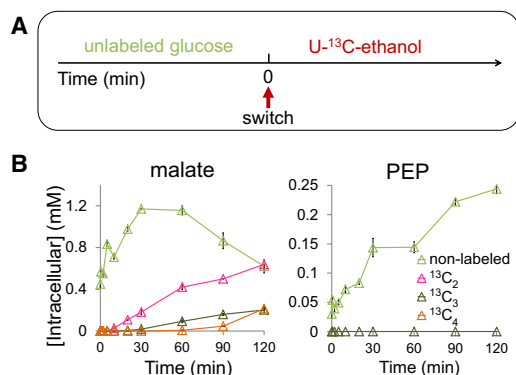


Figure 2. Accumulated PEP Does Not Derive from TCA Cycle Intermediates

(A) Experimental design. Yeast filter cultures (strain Y3358) were grown on 2% unlabeled glucose, and suddenly transferred to 2% U-¹³C-ethanol as the sole carbon source. A various time points thereafter, metabolites were quantitated by LC-MS.

(B) Exogenous ¹³C-ethanol is incorporated into TCA intermediates but not PEP. The x axis represents minutes after the switch, and the y axis represents absolute intracellular concentration (mean ± range of n = 2 biological replicates).

kinase. To evaluate the possibility that PKA inactivation might be necessary upon glucose removal for pyruvate kinase inactivation, we used a yeast strain in which the PKA regulatory subunit Bcy1 is deleted, resulting in constitutive PKA activity. The *bcy1* strain grows normally on glucose but fails to grow on gluconeogenic carbon sources (Toda et al., 1987). Upon glucose removal, however, like wild-type yeast, the *bcy1* strain accumulates PEP (Figure 3). Thus, PKA inactivation is not necessary for acute inactivation of pyruvate kinase upon glucose removal.

AMP-activated protein kinase, SNF1, is active in the absence of glucose and is required for growth on most carbon sources other than glucose. To evaluate a possible role for SNF1 in pyruvate kinase regulation upon glucose removal, we used a yeast strain, *snf1^{as}*, in which *SNF1* was genetically modified (*snf1^{1132G}*) such that it could be specifically inhibited by the ATP analog mMe-PP1 (Zaman et al., 2009). We hypothesized that activation of SNF1 might be required to inhibit pyruvate kinase upon glucose removal. By adding mMe-PP1 simultaneously with glucose removal in the *snf1^{as}* strain, we were able to test this possibility. We found that PEP accumulated to the same extent in the *snf1^{as}* strain upon glucose removal whether or not mMe-PP1 was added concurrently (Figure 3). This result argues against a role for SNF1-based phosphorylation in regulating pyruvate kinase.

Pyruvate Kinase Phosphorylation Is Not Required for Its Regulation

To investigate further a potential role of phosphorylation on pyruvate kinase activity, we inserted a polyhistidine tag at the N terminus of the single genomic *CDC19* locus to enabled efficient purification of the endogenous enzyme. Yeast encoding the tagged pyruvate kinase showed a normal metabolomic response to glucose removal (Figure S3). The abundance and covalent modifications of the tagged enzyme were analyzed in cells growing on glucose and 15 min after glucose removal by

purification and digestion of Cdc19, followed by phosphopeptide enrichment and LC-MS/MS analysis of the resulting peptide fragments. Quantitative comparison of the phosphopeptides under the two conditions was achieved by mixing cells growing exponentially on glucose and ¹⁵N-ammonium sulfate with an equal number of glucose-starved cells fed unlabeled ammonium sulfate. Analysis of both tryptic and Lys-C fragments by LC-MS/MS yielded coverage of the entire protein sequence. Three phosphorylation sites were identified (S9, T21, and S22), as well as previously unknown enzyme acetylation at K475. The T21 phosphorylation was low abundance, with the LC-MS/MS ion count for the phosphopeptide much less than for the unmodified peptide, even after phosphoenrichment. The acetylation and phosphorylation of S22 did not change appreciably after glucose removal. We did observe, however, a 30% decrease in phosphorylation of T21 and an 80% decrease in phosphorylation of S9 in the glucose-starved cells (Figure 4A).

To examine a potential role for the observed phosphorylation in regulating pyruvate kinase, we mutated the genomic copy of *CDC19* to convert S9, T21, or S22 individually to either alanine (A) or glutamate (E), in an effort to mimic a perpetually dephosphorylated or phosphorylated protein, respectively. We then analyzed the growth rate and the metabolome of each of the resulting strains before and after acute glucose removal. The *cdc19^{T21E}* variant (designed to mimic the phosphorylated enzyme, which is low abundance but somewhat more prevalent in glucose-fed cells) resulted in slow growth and elevated PEP even prior to glucose removal, due to constitutively impaired pyruvate kinase activity (see below). Each of the other strains grew normally and, like the wild-type strain, accumulated PEP upon glucose removal, consistent with these phosphorylation sites playing a minimal role in regulating Cdc19 activity (Figure 4B).

Ultrasensitive Allostery Is Sufficient to Modulate Pyruvate Kinase Activity

Having shown that pyruvate kinase regulation does not depend on phosphorylation, we re-examined the allosteric regulation of pyruvate kinase enzyme activity. A simplified mathematical model for glycolysis indicates that in the absence of other regulatory mechanisms, FBP must activate Cdc19 in a cooperative fashion, i.e., with a Hill coefficient of greater than 1, for PEP to accumulate when FBP falls (Figure S4 and the Supplemental Experimental Procedures). However, prior studies indicated that FBP activates Cdc19 but not in a cooperative manner, i.e., with a Hill coefficient approximately equal to 1. These prior in vitro studies, however, measured enzymatic activity with FBP and PEP at nonphysiological concentrations (Mesecar and Nowak, 1997a, 1997b; Bollenbach and Nowak, 2001; Susan-Resiga and Nowak, 2004). While the cellular concentrations of PEP range from 0.03 to 0.8 mM, PEP was typically added in saturating concentrations of 20 mM to facilitate the enzyme activity measurements.

We developed a sensitive, mass spectrometry-based assay for pyruvate kinase activity in order to examine the effect of FBP activation in the presence of PEP at physiological concentrations (Figure 5A). We found that FBP is an ultrasensitive activator with Hill coefficients of 2.4 and 1.6 in the presence of 0.03 and 0.8 mM PEP, respectively, which correspond to the

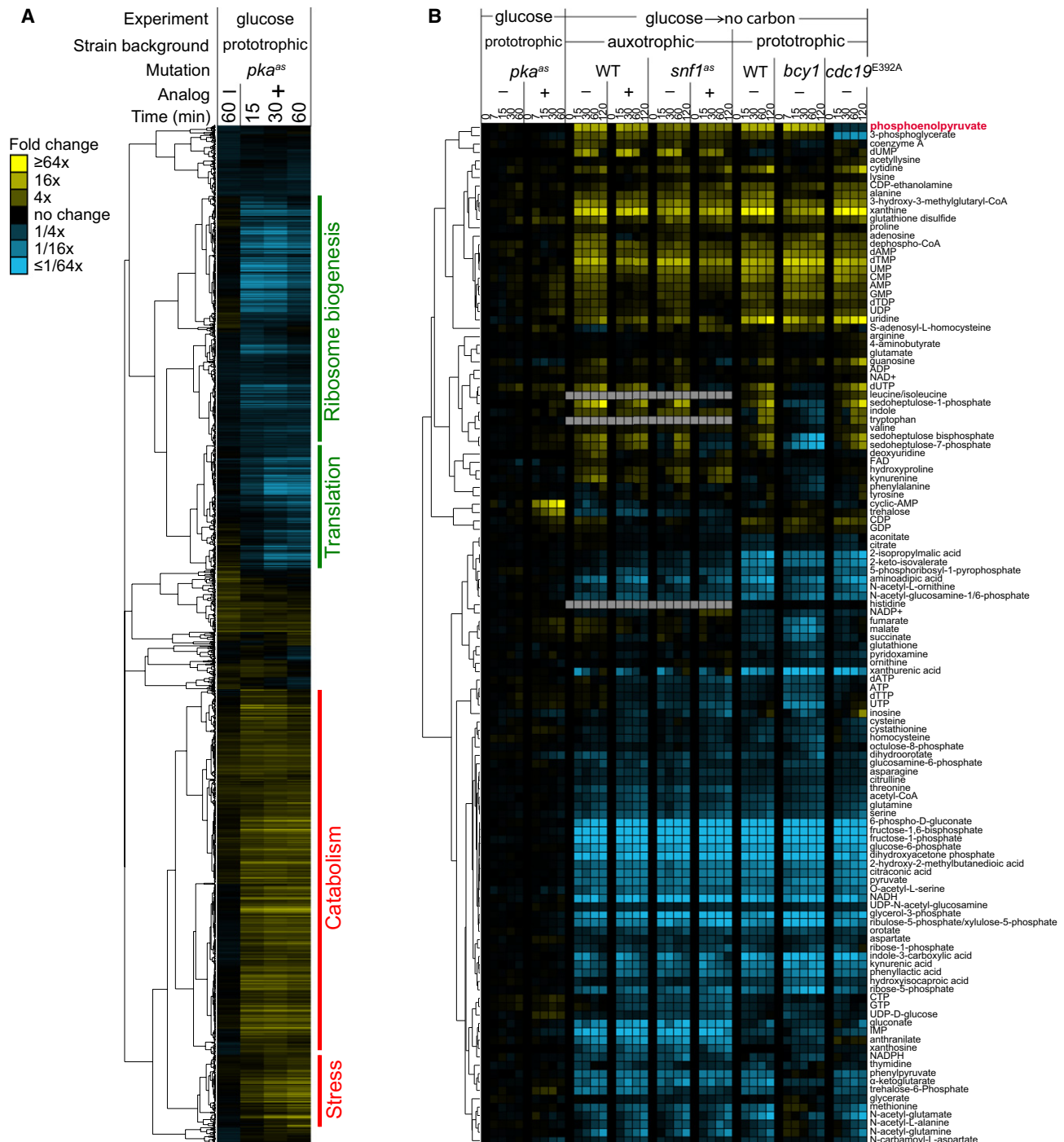


Figure 3. Transcriptome and Metabolome Response to Inhibition of Glucose-Sensing Kinases

Transcriptome (A) and metabolome (B). Note that PEP levels are robust to manipulation of glucose-signaling kinases but sensitive to a point mutation of Cdc19 that ablates allosteric activation by FBP (*cdc19^{E392A}*). In the columns marked “glucose,” transcripts and metabolites of yeast cells growing in minimal liquid media containing 2% glucose were measured during exponential phase starting at A_{600} of 0.4. When indicated, 1 μ M analog 1NM-PP1 was added at time 0 to deactivate genetically engineered PKA (*pka^{as}*). In the columns marked “glucose $\rightarrow$ no carbon,” cells were switched to no carbon, and metabolome was quantitated by LC-MS. When indicated, 1 μ M analog mMe-PP1 was added at time 0 to deactivate genetically engineered SNF1 (*snf1^{as}*). Higher analog concentrations (2.5 μ M and 25 μ M) were also tested, with identical results observed. All experiments in this figure were performed with liquid culture, and all reported values are $\log_2$ transformed fold changes relative to time 0; all metabolites data are mean of duplicate samples at each time point. Gray boxes indicate metabolites that were added to media as nutrients, precluding measurement of their intracellular levels. Strains: Y2864, auxotrophic WT; Y3561, *pka^{as}*; Y3504, *snf1^{as}*; Y3358, prototrophic WT; Y3898, *cdc19^{E392A}*. See also Figure S2.

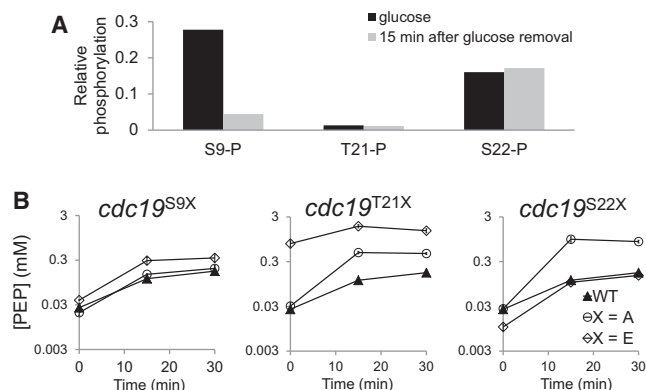


Figure 4. PEP Accumulation upon Glucose Removal Does Not Require Cdc19 Covalent Modification

(A) Cdc19 phosphorylation before and after glucose removal. The relative extent of phosphorylation is defined as (ion count for the phosphopeptide) / (ion count for the phosphopeptide + ion count for its unphosphorylated form), with ion counts for phosphopeptides measured in phosphoenriched samples and those for unphosphorylated peptides in unenriched samples. Due to differences in enrichment and ionization efficiency, this ratio provides only qualitative information regarding the stoichiometry of the phosphorylation events. Differences before and after glucose removal reflect isotope ratio-based measurements and are therefore quantitative.

(B) Mutation of the phosphorylation sites on the genomic copy of Cdc18 to alanine or glutamate does not prevent PEP accumulation upon glucose removal. The x axis represents minutes after glucose removal, and the logarithmic y axis represents absolute intracellular PEP concentration (mean $\pm$ range of $n = 2$ biological replicates).

See also Figure S3.

intracellular PEP concentration of cells growing on glucose and 30 min after being switched to no carbon medium. In contrast, a saturating PEP concentration of 20 mM renders the effect of FBP noncooperative, with a Hill coefficient of 1.0. Thus, in the presence of physiological substrate concentrations, FBP cooperatively activates Cdc19, with Cdc19 activity changing by $>1,000$ -fold over the physiological FBP concentration range.

PEP Accumulation Requires Allosteric Regulation of Pyruvate Kinase

As noted above, yeast cells expressing variants of Cdc19 that mimic S9, T21, or S22 in a phosphorylated or dephosphorylated state still exhibited a significant increase in PEP levels upon glucose removal. This would suggest that each of the mutant enzymes should, like the native enzyme, exhibit cooperative activation by FBP. We purified each of the mutant enzymes from yeast and determined their enzymatic properties. As predicted, each variant enzyme exhibited a Hill coefficient for FBP activation of between 1.6 and 2.0 (Figure 5B). The only mutant with a significant enzymatic defect was T21E, which exhibited persistently low enzyme activity and imparted impaired growth on glucose. Since T21E mimics the phosphorylated enzyme, which is more abundant in the presence of glucose, this low activity cannot account for the inhibition of pyruvate kinase upon glucose removal.

To determine the physiological significance of pyruvate kinase regulation by FBP, we examined the in vitro and in vivo proper-

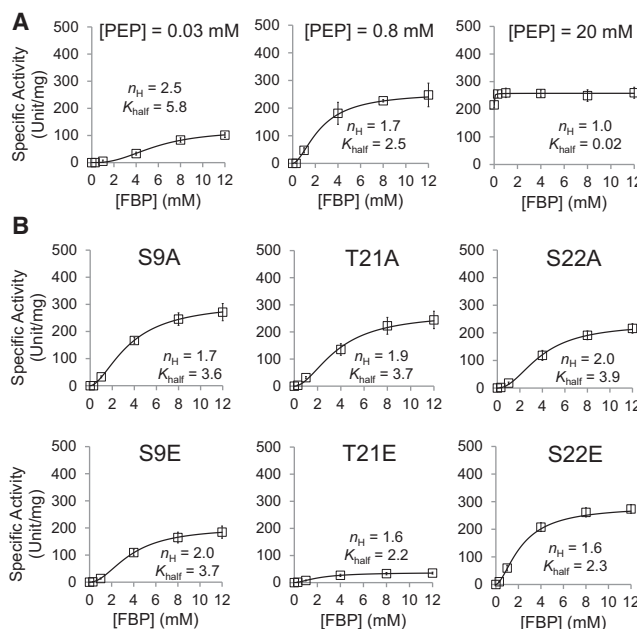


Figure 5. Pyruvate Kinase Exhibits Cooperative Activation by FBP Independent of Its Phosphorylation Status

(A) Activity of purified His-tagged Cdc19 as a function of FBP concentration. The three small plots differ in terms of PEP concentration added, increasing from 0.03 mM (left-most) to 20 mM (right-most).

(B) Activity of Cdc19 variants in which the indicated phosphorylation site has been changed to alanine or glutamate in the presence of 0.8 mM PEP. The x axis represents FBP concentration, and the y axis represents specific pyruvate kinase activity in enzyme activity units (μ mol of product produced per minute) per mg of enzyme. Experimental data (mean $\pm$ range of $n = 2$) were fit to Hill equations (lines). Numbers in the plots are the Hill coefficient (n_H), and the FBP concentration producing half-maximal activation (K_{half}).

See also Figure S4.

ties of a pyruvate kinase mutant desensitized to FBP activation. Fenton and Blair created a number of mutant variants of yeast pyruvate kinase carrying single amino acid changes of the sites likely, based on the enzyme's structure, to participate in FBP allosteric activation (Fenton and Blair, 2002; Friesen et al., 1998). One such mutant enzyme, *cdc19*^{E392A}, appeared to diminish the requirement for FBP for enzymatic activation without affecting either FBP or PEP binding. Using our assay, we confirmed that *cdc19*^{E392A} exhibited full enzymatic activity at physiological concentrations of PEP even in the absence of FBP (Figure 6A). Accordingly, we constructed a strain in which the genomic *CDC19* was mutated to convert E392 to alanine. The resulting cells (*cdc19*^{E392A}) grew normally on glucose but failed to show an increase in PEP levels upon glucose removal (Figure 6B). Thus, cells expressing the FBP-insensitive E392A mutant exhibited inappropriate pyruvate kinase flux, thereby depleting PEP after glucose removal.

Gluconeogenic Growth Requires Pyruvate Kinase Allosteric Regulation

Upon glucose depletion in the presence of a nonfermentable carbon source, *S. cerevisiae* transitions into gluconeogenesis. When that carbon source is ethanol, PEP carboxykinase uses

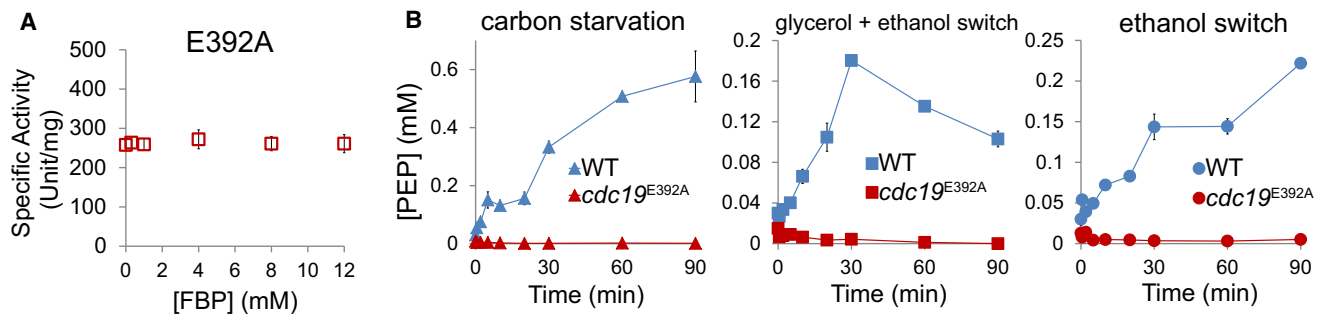


Figure 6. Allosteric Regulation of Cdc19 by FBP Is Essential for Glucose Removal-Induced PEP Accumulation

(A) The Cdc19 E392A variant is active in the absence of FBP. Pyruvate kinase activity was measured in the presence of 0.8 mM PEP and varying concentrations of FBP as per Figure 5.

(B) Genomic substitution of wild-type pyruvate kinase with E392A (*cdc19*^{E392A}) eliminates PEP accumulation upon glucose removal. The x axis represents minutes after the indicated switch, and the y axis represents absolute intracellular PEP concentration (mean $\pm$ range of N = 2 biological replicates). Experiments were performed with filter culture.

See also Figure S5.

ATP to drive the synthesis of PEP from oxaloacetate, whose production from pyruvate also requires ATP. If pyruvate kinase were simultaneously active, ATP would be consumed in a futile cycle. However, even though Cdc19 protein is present during exponential gluconeogenic growth at $\sim 20\%$ of its level in glucose-fed cells, futile cycling is undetectable (Cortassa et al., 1995; Wilson and Bhattacharjee, 1986). Because gluconeogenic growth is associated with reduced FBP levels, we hypothesized that one physiological function of the regulation of pyruvate kinase by FBP is to prevent futile cycling during gluconeogenesis.

To explore this hypothesis, we transferred wild-type and *cdc19*^{E392A} yeast from glucose-containing media to minimal liquid media containing ethanol as the sole carbon source. Upon sudden glucose removal, wild-type cells initiated gluconeogenic growth after ~ 1 day of lag phase. In contrast, *cdc19*^{E392A} cells did not start growing until two weeks later, and those cells that eventually grew out showed evidence of suppressor mutations, i.e., they quickly initiated gluconeogenesis in repeat experiments (Figure S5A). Yeast cells expressing the Cdc19 variants that mimic S9, T21, and S22 in a phosphorylated or dephosphorylated state exhibited normal initiation of gluconeogenesis and identical growth on ethanol compared to the wild-type cells (Figure S5B). These results are consistent with a primary role of allosteric regulation in controlling futile cycling at the glycolysis-TCA cycle interface during gluconeogenic growth.

Pyruvate Kinase Allosteric Regulation Improves Energy Charge Homeostasis upon Glucose Upshift

The FBP-insensitive mutant also allowed us to test the physiological function of the accumulation of PEP and other lower glycolytic intermediates after glucose removal. We hypothesized that the accumulated lower glycolytic intermediates might serve as a reservoir of readily accessible high energy phosphate bonds upon glucose addition. Such a reservoir could be physiologically valuable, as glucose upshift of yeast results in a marked drop in ATP concentration, since phosphorylation of the newly available glucose rapidly consumes ATP (Walther et al., 2010).

To test the possibility that Cdc19 allosteric regulation contributes to energy homeostasis upon glucose addition, we added

glucose to glucose-starved wild-type and *cdc19*^{E392A} yeast. In wild-type cells, glucose addition resulted in a transient decrease in ATP and energy charge, accompanied by the consumption of pre-existing PEP within in 20 s, coincident with the increase in FBP (Figures 7A and 7B and Figures S6A and S6B). In the *cdc19*^{E392A} mutant, a significant pre-existing PEP pool did not exist, and ATP and adenylate energy charge dropped more severely and recovered more slowly.

To mitigate the decrease in energy charge upon glucose upshift, yeast cells activate the enzyme AMP deaminase, Amd1. Amd1 converts AMP into IMP, which can be later recycled into AMP (Walther et al., 2010). Consistent with defective energy charge management, *cdc19*^{E392A} cells accumulate more IMP than do wild-type cells. To explore the possibility that pyruvate kinase allosteric regulation and Amd1 work in concert to maintain energy charge during glucose upshift, we examined energy management in both wild-type and *cdc19*^{E392A} strains from which we had deleted *AMD1*. The *cdc19*^{E392A}*amd1* strain manifested a large accumulation of AMP and a significantly greater drop in energy charge than that in either *cdc19*^{E392A} or *amd1* (Figure 7C and Figure S6C). Thus, we conclude that allosteric regulation of pyruvate kinase plays a significant role in energy homeostasis in yeast.

To test the functional significance of the impaired energy charge homeostasis during glucose upshift, we grew wild-type, *cdc19*^{E392A}, *amd1*, and *cdc19*^{E392A}*amd1* yeast on steady versus oscillating glucose. While the point mutation in Cdc19 had no impact on growth rate in steady glucose, *cdc19*^{E392A} strains grew more slowly on oscillating glucose (Figures 7D–7F, Figures S6D and S6E, and Movie S1). Thus, ultrasensitive regulation of pyruvate kinase facilitates rapid cellular adaptation to varying glucose availability, including rapid resumption of growth upon glucose upshift.

DISCUSSION

Both allosteric activation and covalent modification by phosphorylation have been implicated in regulation of pyruvate kinase, which catalyzes the main efflux from glycolysis. Here

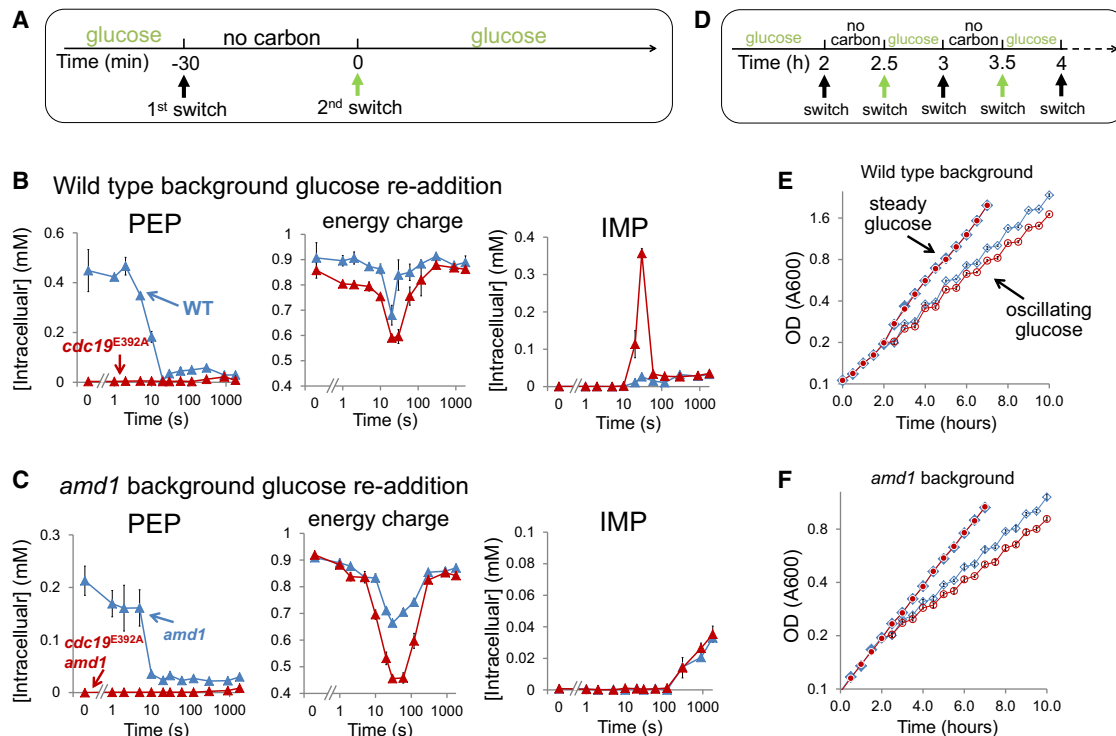


Figure 7. Cdc19 Allosteric Regulation Facilitates Energy Charge Homeostasis upon Glucose Upshift and Enhances Growth on Oscillating Glucose

(A) Experimental design for (B) and (C). Filter cultures growing on glucose were switched to no-carbon media for 30 min. Thereafter, cells were switched to glucose and metabolome was quantified.

(B) [PEP], energy charge = $([ATP] + 0.5 [ADP]) / ([ATP] + [ADP] + [AMP])$, and [IMP] upon glucose readdition in wild-type and *cdc19^{E392A}* yeast. The logarithmic x axis represents seconds after glucose readdition, and the y axis represents absolute intracellular metabolite concentration (mean $\pm$ range of $n = 2$ biological replicates).

(C) Same as (B) but for *amd1* and *cdc19^{E392A}amd1* strains. Amd1 catalyzes conversion of AMP to IMP.

(D) Experimental design for (E) and (F). Filter cultures were alternated between glucose and no-carbon media every 30 min for ~ 10 hr, and growth was monitored.

(E) Growth of wild-type and *cdc19^{E392A}* yeast on steady versus oscillating glucose. Closed symbols, steady glucose; open symbols, oscillating glucose; blue, wild-type; red, *cdc19^{E392A}*. The x axis represents time in hours, and the logarithmic y axis represents optical density (A_{600}) (mean $\pm$ range of $n = 2$ biological replicates).

(F) Same as (E) but for *amd1* (Y3996) and *cdc19^{E392A}amd1* (Y3997).

See also Figure S6 and Movie S1.

we have examined how each of these regulatory processes contributes to flux control through the main yeast pyruvate kinase isozyme, Cdc19. We provide evidence that covalent modification of Cdc19 does not contribute significantly to acute regulation of flux, whereas ultrasensitive allostery provides the dominant flux control mechanism. This ultrasensitive allostery enables switch-like inhibition of pyruvate kinase during glucose removal (Figures 5A and 6) and rapid (less than 5 s) reactivation of the enzyme during glucose readdition (Figure 7B). From a metabolic standpoint, this switch-like behavior results in accumulation of three carbon units and high energy phosphate bonds in the glycolytic pathway after glucose removal. This accumulation facilitates the onset of gluconeogenesis (Figure S5) and provides a boost to reinitiating fermentative growth when glucose becomes available (Figure 7).

An advantage of allosteric regulation over phosphorylation is speed (Kern and Zuiderweg, 2003). In the case of Cdc19, such speed, in combination with ultrasensitivity, enables pyruvate

kinase inactivation to outpace depletion of upstream glycolytic intermediates, resulting in the seemingly paradoxical accumulation of PEP upon glucose removal. A similar accumulation of PEP upon glucose removal also occurs in prokaryotes, where PEP is the substrate for glucose import and phosphorylation. Accordingly, PEP accumulation is required to enable rapid assimilation of glucose when it becomes available. Here we show that PEP and phosphoglycerate accumulation in yeast plays a related role, acting as a store of high-energy phosphate bonds that helps to maintain energy charge upon glucose upshift.

The contrast in dynamics between activation of pyruvate kinase by FBP and activation of phosphofructokinase by fructose-2,6-bisphosphate is informative. Fructose-2,6-bisphosphate levels are regulated through cyclic-AMP-based signaling, which occurs on the timescale of minutes (dos Passos et al., 1992). In contrast, the turnover of glycolytic intermediates occurs in seconds. Thus, phosphofructokinase is designed to turn off more slowly than pyruvate kinase upon glucose removal. Such

timing traps residual glucose-derived carbon in a high energy form as lower glycolytic intermediates.

The ultrasensitive activation of pyruvate kinase by FBP is a striking example of intrinsic metabolic regulation: control of metabolic flux solely by enzymes and core metabolites, without reliance on transcription, covalent modification, or small molecule signals. The evolution of such intrinsic regulation is not surprising, given the centrality of pyruvate kinase regulation to the switch between glycolysis and gluconeogenesis, an ancient biological challenge. More intriguing is the fact that allostery has persisted as the dominant regulatory mechanism for yeast pyruvate kinase, and perhaps more generally for microbial metabolism.

The predominance of pyruvate kinase regulation by allostery, despite the enzyme being covalently modified by phosphorylation and acetylation, raises a more general question (Huttlin et al., 2010; Rikova et al., 2007; Wang et al., 2010; Zhao et al., 2010): to what extent is covalent modification functionally significant? Over 4,000 protein phosphorylation sites have been identified in yeast (Bodenmiller et al., 2010; Ptacek et al., 2005). These include PKA sites on ~45 metabolic enzymes, e.g., phosphofructokinase and pyruvate kinase in glycolysis, pyruvate carboxylase in anaplerosis, PEP carboxykinase in gluconeogenesis, acetyl-CoA oxidase in fatty acid beta oxidation, glycogen synthase and phosphorylase in glycogen metabolism, phosphatidylserine synthase and choline kinase in phospholipids biosynthesis, and trehalase in trehalose metabolism (Ptacek et al., 2005). Here we find, however, that the primary soluble metabolic target of PKA is trehalose. While we cannot rule out a subtle role of PKA in nontranscriptional regulation of other metabolic pathways, we believe that this finding is a harbinger of a broader paradigm—most protein covalent modification events, at least in microbes, will prove to have minimal functional impact, being neither selected for or against evolutionarily. This in no way undermines the importance of covalent modification as a regulatory mechanism, but merely suggests that biochemical promiscuity may mask functional specificity. Identifying the functionally important events—which in metabolism can be achieved using the combination of metabolomics and genetics employed here—holds the potential to simplify and clarify the signaling-metabolism interface, and perhaps signaling systems in general.

EXPERIMENTAL PROCEDURES

Yeast Strains

Yeast strains and primers are provided in Tables S1 and S2, and their construction is described in the Supplemental Experimental Procedures.

Yeast Culture Conditions and Extraction

The metabolome of *Saccharomyces cerevisiae* was characterized in cells growing either in liquid batch cultures or on filters on top of an agarose-media mixture. The filter culture technique allows rapid, nondisruptive glucose removal, isotope labeling, and metabolome harvesting by transfer to cold organic solvent (40:40:20 acetonitrile:methanol:water at -20°C) (Rabinowitz and Kimball, 2007). This method was complemented by metabolome harvesting by fast filtration of liquid cultures. The liquid cultures were necessary for the experiments involving analog treatment (1NM-PP1 and mMe-PP1), as diffusion of analogs through the agarose-media and filter to the cells was unreliable. For details on culture conditions and metabolome sampling, see the Supplemental Experimental Procedures.

Metabolite Measurement

Cell extracts were analyzed by reversed phase ion-pairing liquid chromatography (LC) coupled by electrospray ionization (ESI) (negative mode) to a high-resolution, high-accuracy mass spectrometer (Exactive; Thermo Fisher Scientific) operated in full scan mode at 1 s scan time, 10^5 resolution, with compound identities verified by mass and retention time match to authenticated standard (Lu et al., 2010). For a list of metabolites, exact masses, and retention times, see Table S3. Confirmatory analysis for a subset of compounds was conducted by a different reversed phase ion-pairing LC method coupled by negative mode ESI to a Thermo TSQ Quantum triple quadrupole mass spectrometer operating in multiple reaction monitoring mode (Lu et al., 2008) (see Table S4). All data reported in the paper is from the high-resolution LC-MS analysis. Isomers are reported separately only where they fully chromatographically resolved. Absolute intracellular metabolite concentrations in steadily growing *S. cerevisiae* were determined as described previously (Bennett et al., 2008). Metabolite concentrations after perturbations were computed based on the fold change in ion counts relative to steadily-growing cells (grown and analyzed in parallel) multiplied by the absolute concentration in the steadily growing cells as measured with an isotope ratio-based approach.

Measurements of Pyruvate Kinase Activity

S. cerevisiae pyruvate kinase used in biochemistry assays was purified from yeast that carried vector encoding His-tagged wild-type or altered Cdc19. Activity was measured based on LC-MS analysis of the reaction product ATP, instead of coupling of pyruvate kinase reaction to lactate dehydrogenase reaction and measurement of the decrease of NADH at 340 nM with a spectrophotometer. This avoids artifacts attributable to contaminants in purified pyruvate kinase that slowly oxidize NADH. This issue becomes important when quantifying the low pyruvate kinase activity found in low FBP conditions. For further information, see the Supplemental Experimental Procedures.

LC-MS/MS Analysis of Yeast Pyruvate Kinase Protein

Liquid cultures of *S. cerevisiae* strains with His-tagged CDC19 on the chromosome were grown in glucose minimal media containing 5 g/liter unlabeled or ^{15}N -labeled $(\text{NH}_4)_2\text{SO}_4$. Cultures grown in ^{15}N - $(\text{NH}_4)_2\text{SO}_4$ were harvested at a density of A600 = 0.6 by pouring of cell cultures into precooled -20°C ice. Cells growing in unlabeled $(\text{NH}_4)_2\text{SO}_4$ were subjected to carbon starvation at A600 = 0.6, and harvested 15 min after the removal of glucose. Unlabeled and ^{15}N -labeled cells were mixed and ground with Cryo Mill (Retsch, Newtown, PA) at -196°C , and pyruvate kinase was purified by nickel column affinity chromatography (ProBond Purification Systems, Invitrogen).

Purified pyruvate kinase samples were subjected to trypsin or Lys-C digestion. Peptides were then directly analyzed or subjected to titanium dioxide phosphopeptide enrichment and analyzed by a reverse-phase nano-LC-MS/MS. The resulting ion counts for the phosphopeptides in the phosphoenriched sample and for the unphosphorylated peptides in the nonphosphoenriched sample were used in the rough comparison of the stoichiometry of the phosphorylation events. For details, see the Supplemental Experimental Procedures.

ACCESSION NUMBERS

Microarray data in this study can be viewed and downloaded from the PUMAdb website at http://puma.princeton.edu/cgi-bin/publication/viewPublication.pl?pub_no=544.

SUPPLEMENTAL INFORMATION

Supplemental Information includes six figures, five tables, Supplemental Experimental Procedures, and one movie and can be found with this article online at <http://dx.doi.org/10.1016/j.molcel.2012.07.013>.

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